Predicting Bus Stop Passenger Volume in Singapore from Spatial and Network Features

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Abstract

We predict passenger volume at Singapore bus stops from spatial and network features derived from public datasets on residential density, MRT connectivity, points of interest, and bus-line structure. Using engineered per-stop features, we compare linear and tree-based models across four temporal targets (weekday morning, midday, evening, and weekend/holiday averages) under repeated train/test splits.

1. Introduction

Public bus networks carry the majority of public-transport trips in Singapore, and understanding where ridership concentrates is central to operational and planning decisions: where to add capacity, where to open new interchanges, and where to consolidate under-used stops. Passenger counts are expensive to collect directly, but a large amount of information about a stop's surroundings is freely available — residential density, proximity to rail, nearby amenities, and the stop's role in the bus network itself. It is therefore useful to know how well such features, on their own, predict the volume a stop will see.

We treat this as a supervised regression problem. For each stop we construct thirteen features from five public datasets and predict four summary volumes per stop: mean hourly boardings-plus-alightings on weekday mornings, weekday middays, weekday evenings, and weekend/holiday daytimes. The four targets let us ask a secondary question: do the same features drive ridership in different parts of the week, or does the relationship change between commute and leisure periods?

The project has three goals, aligned with the questions that a transit operator would naturally ask: (i) *predictive*, which class of models fits this problem best; (ii) *inferential*, which covariates carry the signal and whether the data-generating mechanism looks linear; and (iii) *practical*, whether a simpler, interpretable model is sufficient or whether the accuracy gap to a black-box model is worth the loss of interpretability.

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2. Data

Sources. We use the LTSG dataset (Lan et al., 2022), which bundles five tables relevant to Singapore transit: hourly bus passenger counts (`bus_vol`), bus-line routing (`bus_line`), residential block attributes (`hdb`), MRT station locations (`mrt`), and geocoded points of interest (`poi`). LTSG does not include coordinates for bus stops, so we join them against the LTA Bus Stop dataset on data.gov.sg (Land Transport Authority, 2025). Of ∼5,000 stops in `bus_vol`, 23 ($< 0.5\%$) have no match in the current LTA data and are dropped; this is plausibly due to stops being retired between the two datasets' collection periods. After the join and requiring complete targets (see below), $n = 4985$ stops remain.

Target construction. The raw `bus_vol` table gives hourly tap-in (`in`) and tap-out (`out`) counts per stop, labelled weekday (WD) or holiday/ weekend (H). Visualisation of the hourly distribution (Figure 1) shows two clear phenomena: a right-skewed distribution across stops at every hour, and sharply different temporal patterns on weekdays versus weekends. We therefore define volume as in + out and construct four targets per stop by averaging volume over four hour ranges:

- `wd_am`: weekday, 06:00–08:59 (morning peak);
- `wd_pm`: weekday, 16:00–19:59 (evening peak);
- `wd_midday`: weekday, 09:00–15:59 (off-peak);
- `h_avg`: weekend/holiday, 06:00–22:59 (daytime).

The evening peak is extended to 19:59 to reflect later typical working hours locally. All four are continuous scalars, so the supervised task is regression.

Feature construction. We engineer $p = 13$ features per stop, all computable from the stop's coordinates and the five source tables. Spatial features use a Haversine distance and

Table 1. Per-stop features. All counts and sums are computed within the stated radius of the stop. `is_terminal` is binary; all other features are non-negative reals or counts.

Name	Source	Description
`hdb_units_500m`	HDB	dwelling units
`hdb_rentals_500m`	HDB	rental units
`hdb_commercial_500m`	HDB	commercial blocks
`dist_nearest_mrt`	MRT	metres to nearest station
`mrt_within_1km`	MRT	MRT stations within 1 km
`poi_education_500m`	POI	schools, universities
`poi_food_500m`	POI	restaurants, cafes, bars
`poi_health_500m`	POI	clinics, hospitals
`poi_leisure_500m`	POI	parks, museums, attractions
`poi_shopping_500m`	POI	retail, malls, supermarkets
`poi_worship_500m`	POI	religious sites
`n_lines`	bus_line	distinct bus lines at stop
`is_terminal`	bus_line	first/last stop on any route

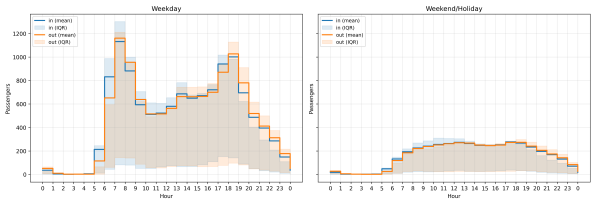

Figure 1. Mean hourly tap-in (blue) and tap-out (orange) across stops, with shaded IQR bands, on weekdays (left) and weekends/holidays (right). Weekday peaks are pronounced and directionally asymmetric; weekend volume is flatter and smaller. Motivates the four-target split.

are computed against stop centroids. Table 1 summarises the feature set; rationale for each is given in the EDA notebook. Radii (500 m for HDB and POI, 1 km for MRT) reflect typical walking distances in Singapore: longer for rail because riders walk further to stations.

Missingness. There are no null values in the raw feature tables. However, `bus_vol` does not record zero-volume hours explicitly — a stop with no activity during a given hour simply has no row for that cell. A small number of stops are missing observations for at least one target bucket — 15 stops are missing weekday midday data, 13 are missing weekday morning, and a handful have gaps in the other buckets. These partial rows are dropped, leaving $n = 4985$ stops with complete targets for modelling.

Subsetting. No further subsetting is applied. The dataset size (a few thousand stops with thirteen features) is well within what all methods in the course handle without downsampling.

3. Methods

We compare a set of regression models chosen to span the main mechanisms the course covered: pure linear, regularised linear, and two tree ensembles. Running a range from rigidly linear to flexibly nonlinear lets us see directly how much structure the data supports.

Models.

- **Linear regression** (OLS). Baseline; gives a reference for how much of the variance is explained by simple additive effects.
- **Ridge regression** with ℓ_2 penalty. Guards against ill-conditioning from correlated features (e.g. HDB units and rentals) and provides a fair linear benchmark.
- **Random forest**. Flexible nonlinear, captures interactions. Standard black-box reference.
- **Gradient boosted trees**. Typically the strongest tabular baseline; tests whether an ensemble boosting approach materially beats a random forest here.

Each model is fit independently on each of the four targets; we do not attempt joint multi-output training (this is discussed as a limitation in Section 6).

Evaluation protocol. We report 5-fold cross-validated performance, which gives a distribution of five test scores per model per target rather than a single number. Folds are shuffled once with a fixed seed, and the same fold indices are used for every model, so pairwise comparisons are on matched test sets. The primary metric is test RMSE on the natural passenger-count scale; mean absolute error (MAE) is reported alongside as a robustness check. Because targets are right-skewed, models are trained on $\log(1 + y)$ and predictions are inverse-transformed before scoring.

Statistical significance of pairwise model differences is assessed with a paired t-test across folds: paired because models are evaluated on identical folds, which removes fold-level variance (some folds contain harder test stops than others). A small p-value indicates consistent outperformance across folds rather than luck on particular splits.

Hyperparameters. Ridge α is selected by `RidgeCV` over 20 values logspaced in $[10^{-2}, 10^{3}]$, using its built-in leave-one-out criterion on the outer training fold. For the tree ensembles, a nested 3-fold cross-validation with `GridSearchCV` selects random forest's `min_samples_leaf` from $\{5, 10, 20\}$, and gradient boosting's `learning_rate` from $\{0.05, 0.1\}$ and `max_depth` from $\{\text{None}, 5, 8\}$. Random-forest tree count is fixed at 200, which is well into the diminishing-returns

Table 2. Test RMSE (mean over 5 folds) by model and target; lower is better. Bold: best per target.

Model	wd_am	wd_pm	wd_midday	h_avg
Linear	4804.4	4667.0	4357.5	1466.8
Ridge	4679.9	4550.3	4201.7	1416.0
Random forest	3232.8	3299.4	2166.9	868.1
Gradient boost	**3079.0**	**3052.3**	**2019.0**	**771.3**

regime for this dataset size; gradient boosting uses the library's default of 100 trees with early stopping, so its effective count is data-dependent.

4. Results

Gradient boosting wins on every target. Table 2 reports test RMSE across all four models and targets. Gradient boosting is the lowest-error model on every target, with random forest a close second, ridge third, and linear regression fourth. The ranking is identical across targets. Relative to the linear baseline, gradient boosting reduces RMSE by 36% on wd_am, 35% on wd_pm, 54% on wd_midday, and 47% on h_avg.

The mechanism is strongly nonlinear. The gap between linear and tree-based models is the dominant effect in the comparison: ∼1500–2300 RMSE points across all four targets, corresponding to a roughly 35–55% error reduction. That the tree models also interact features (gradient boosting and random forest both split on thresholds and combine splits multiplicatively) is consistent with the mechanism being multivariate-nonlinear, not a failure to include the right linear terms.

Interestingly, the paired t-test does *not* declare the tree-vs-linear gaps significant at $p < 0.05$ despite their size (Table 3; p-values range 0.12–0.27). The reason is visible in Figure 2: linear-model RMSE varies enormously across folds (interquartile range ∼2500–6000 on wd_am), which inflates the test statistic's denominator. The size of the gap is real; the test simply lacks power against this much fold-level variance with five folds. We report the raw difference alongside the p-value for this reason.

Ridge barely improves over OLS. Ridge reduces RMSE by only 2–4% across targets, confirming that multicollinearity is not the bottleneck for linear fit. The 13 features come from different source tables and use different radii, so they are reasonably de-correlated by design. Regularisation is not what linear models need here — more flexible functional form is.

Gradient boosting vs. random forest. GradBoost's edge over random forest is small in magnitude (97–247 RMSE) but more consistent across folds and reaches significance on wd_am ($p = 0.003$) and h_avg ($p = 0.027$). For wd_pm and wd_midday the difference does not reach significance. The practical implication is that choice of tree ensemble matters less than the linear-vs-tree decision: either tree-based option closes nearly all of the gap against the linear baseline.

Table 3. Paired t-test comparing gradient boosting (best per target) against each alternative. ΔRMSE is the excess RMSE of the alternative. Bold: significant at $p < 0.05$.

Target	Comparison	ΔRMSE	p-value
wd_am	GB vs. Linear	1725.4	0.258
	GB vs. Ridge	1600.9	0.267
	GB vs. RF	153.8	**0.003**
wd_pm	GB vs. Linear	1614.7	0.259
	GB vs. Ridge	1498.1	0.267
	GB vs. RF	247.1	0.104
wd_midday	GB vs. Linear	2338.5	0.123
	GB vs. Ridge	2182.7	0.131
	GB vs. RF	147.8	0.097
h_avg	GB vs. Linear	695.5	0.160
	GB vs. Ridge	644.7	0.171
	GB vs. RF	96.8	**0.027**

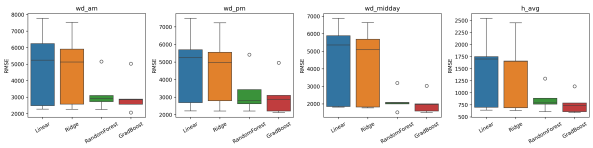

Figure 2. Per-fold test RMSE by model, one panel per target. Tree-based models (random forest, gradient boosting) are consistently far below linear models, but linear-model RMSE also varies dramatically across folds, which explains why the large gap does not reach statistical significance in a paired test.

5. Stop Typology via Clustering

To complement the supervised analysis, we cluster stops on their four-target usage profile (each stop's target vector normalised to sum to 1) and ask whether natural groupings correspond to interpretable transit archetypes. Normalising to profiles separates temporal *pattern* from overall *busyness* — an unnormalised clustering on this data simply ranks stops by total volume, which is not what we want. We fit K-means with $K = 4$, motivated by the regimes visible in the hourly data (weekday morning peak, weekday evening peak, midday off-peak, weekend) plus a natural split between morning-origin and evening-destination stops.

Four archetypes emerge. Table 4 labels the resulting clusters.

Table 4. K-means cluster archetypes ($K = 4$), fit on per-stop normalised target profiles.

k	Profile	Interpretation
0	AM > PM > midday > weekend	General residential
1	Dominant AM, near-zero weekend	Residential origin
2	Dominant PM, near-zero weekend	Workplace destination
3	Even AM / PM / midday, low weekend	Mixed-use

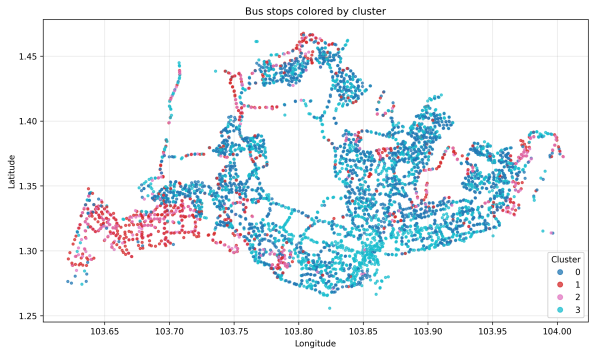

Figure 3. Singapore bus stops coloured by K-means cluster ($K = 4$), fit on normalised target profiles. Morning-origin (red, cluster 1) and evening-destination (pink, cluster 2) stops concentrate in the western Jurong region; balanced (blue, 0) and mixed-use (cyan, 3) archetypes dominate elsewhere.

Geographic coherence. The clustering operates on temporal profiles only — coordinates are not used — yet the archetypes lay out coherently on the map (Figure 3). Morning-origin and evening-destination stops (clusters 1 and 2) concentrate in the western region around Jurong, the part of Singapore with the clearest residential–industrial commute structure: residents tap in during the morning rush at dormitory-style estates and tap out in the evening at workplace clusters. Balanced and mixed-use stops (clusters 0 and 3) dominate elsewhere, including the core and the northeastern residential belt.

No cluster is dominated by weekend ridership; weekend volume is uniformly modest across all archetypes. Temporal variation in this network is primarily a weekday phenomenon, which also frames the supervised result that weekend prediction is the hardest of the four targets.

6. Limitations

Several caveats qualify the findings.

Temporal scope. LTSG is a snapshot from one period; ridership patterns shift with urban development, COVID-era work-from-home adjustments, and new MRT lines. Our features reflect current infrastructure from LTA; a small mismatch between the two periods is unavoidable.

Spatial aggregation choices. The 500 m / 1 km radii are defensible but not optimised.

Feature coverage. We do not use POI quality signals (ratings, price level) or bus-timing columns (first/last bus, which could yield a service-span feature). Including these is a natural extension.

Independence assumption. Our evaluation protocol treats stops as i.i.d. samples, but stops close together share the same local environment and likely have correlated volumes.

Four separate models. Fitting each target independently ignores the clear correlation between, e.g., `wd_am` and `wd_pm`. A multi-output model could share strength and yield tighter estimates, particularly for the smaller weekend signal.

7. Conclusion

Impact Statement

This work uses only public, non-personal data (aggregated passenger counts, infrastructure locations, and commercial amenity metadata). The intended use is operational planning, where accurate demand estimation can improve service efficiency and equity of access; failure modes (e.g. under-serving stops with systematically underestimated volume) are worth checking against ground truth before deployment. No individual-level data is involved.

References

Lan, T., Cheng, H., Wang, Y., and Wen, B. Site selection via learning graph convolutional neural networks: A case study of singapore. *Remote Sensing*, 14(15), 2022. ISSN 2072-4292. doi: 10.3390/rs14153579. URL https://www.mdpi.com/2072-4292/14/15/3579.

Land Transport Authority. LTA bus stop (2026). data.gov.sg, 2025. URL https://data.gov.sg/datasets/d_3f172c6feb3f4f92a2f47d93eed2908a/view. Dataset. Retrieved April 19, 2026.

A. Appendix: Reproducibility

Code and setup instructions are in the project repository. The pipeline is three notebooks: `01_eda.py` (data exploration and target design), `02_features.py` (feature engineering and join), and `03_model.py` (experiments, hyperparameter selection, and plotting). All random seeds are fixed.